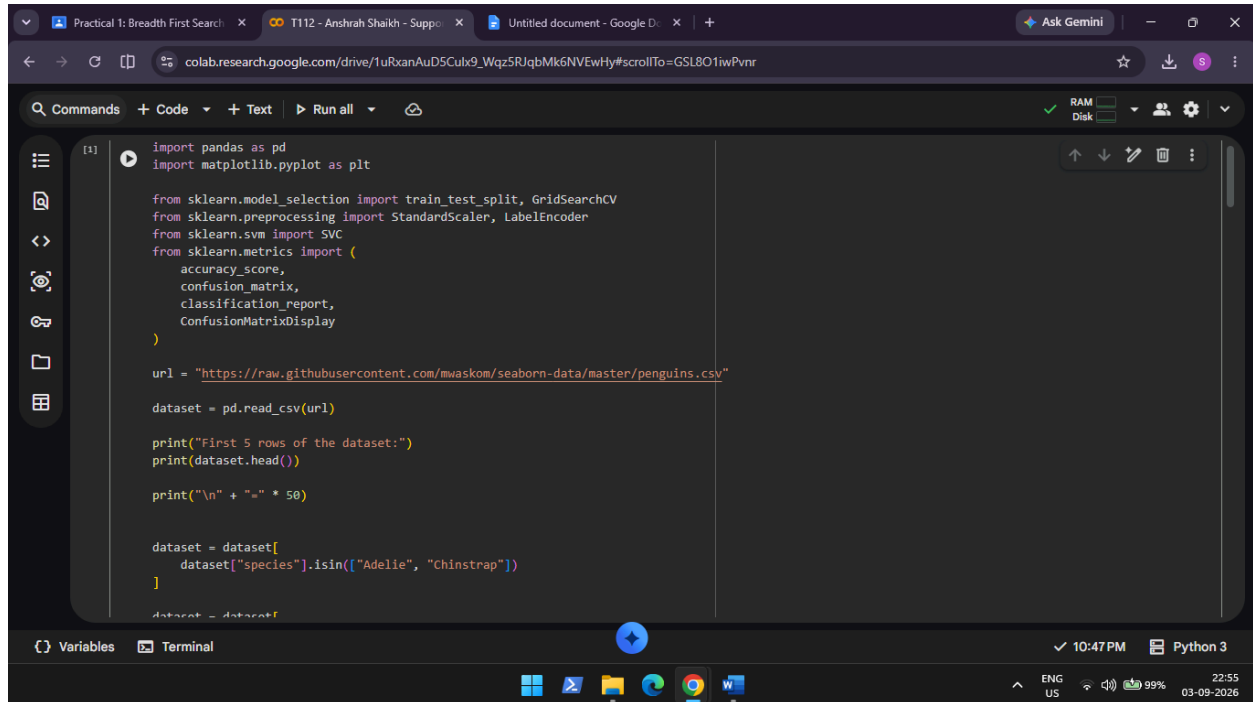


# MVLU COLLEGE

AIM: Support Vector Machines (SVM)

- Implement the SVM algorithm for binary classification.
- Train an SVM model using a given dataset and optimize its parameters.
- Evaluate the performance of the SVM model on test data and analyze the results.

INPUT:



```
[1] import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.svm import SVC
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay
)

url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/penguins.csv"

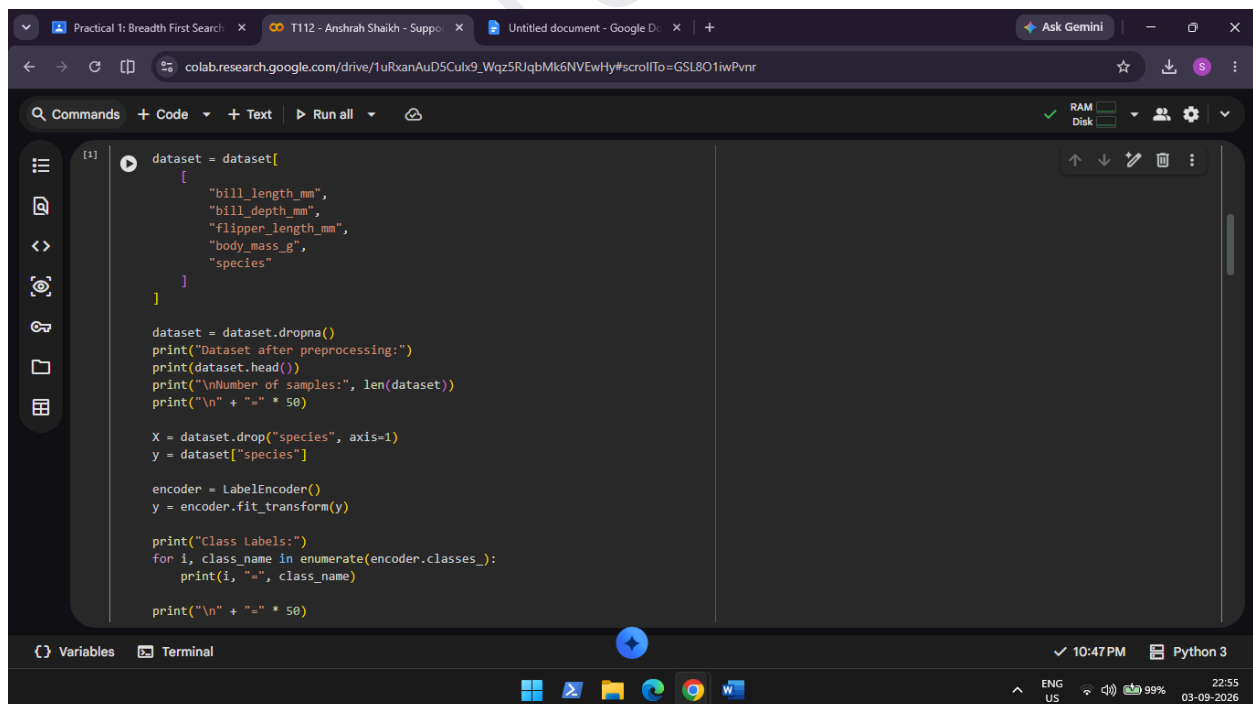
dataset = pd.read_csv(url)

print("First 5 rows of the dataset:")
print(dataset.head())

print("\n" + "=" * 50)

dataset = dataset[
    dataset["species"].isin(["Adelie", "Chinstrap"])
]

dataset = dataset
```



```
[1] dataset = dataset[
    [
        "bill_length_mm",
        "bill_depth_mm",
        "flipper_length_mm",
        "body_mass_g",
        "species"
    ]
]

dataset = dataset.dropna()
print("Dataset after preprocessing:")
print(dataset.head())
print("\nNumber of samples:", len(dataset))
print("\n" + "=" * 50)

X = dataset.drop("species", axis=1)
y = dataset["species"]

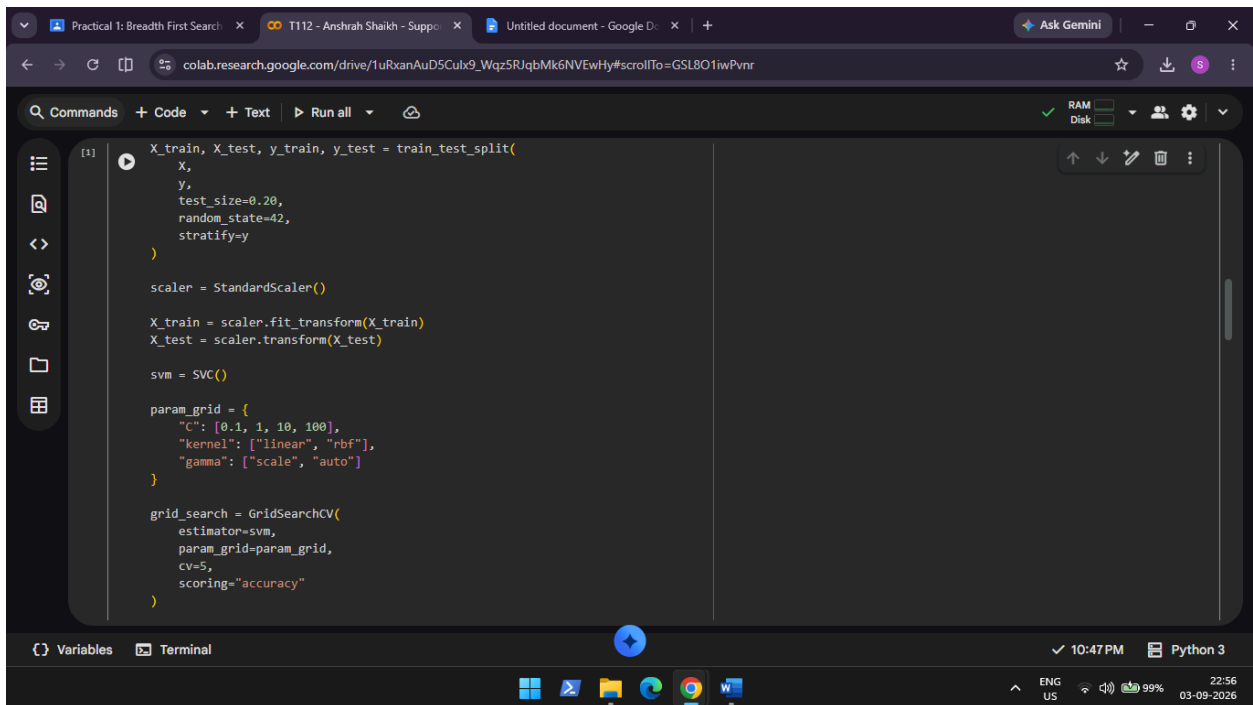
encoder = LabelEncoder()
y = encoder.fit_transform(y)

print("Class Labels:")
for i, class_name in enumerate(encoder.classes_):
    print(i, "=", class_name)

print("\n" + "=" * 50)
```

ANSHRAH SHAIKH  
T112  
TYCS  
AI PRACTICAL 5

# MVLU COLLEGE



```
[1] X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

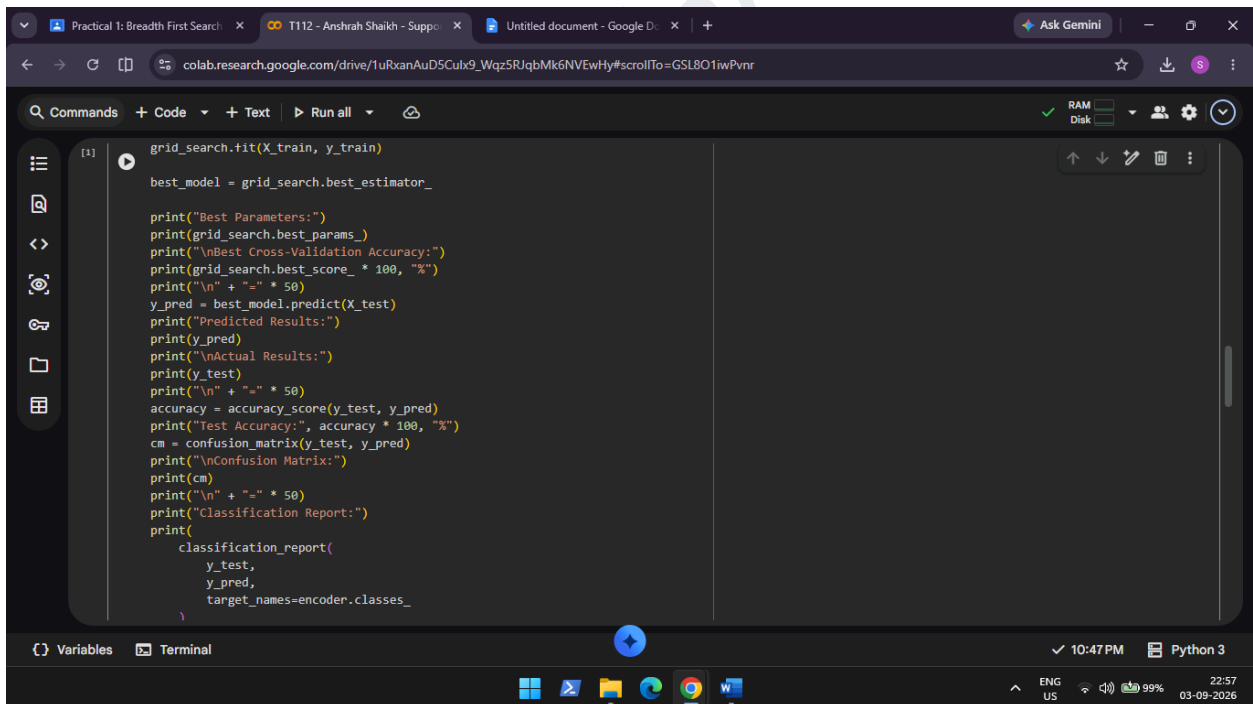
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

svm = SVC()

param_grid = {
    "C": [0.1, 1, 10, 100],
    "kernel": ["linear", "rbf"],
    "gamma": ["scale", "auto"]
}

grid_search = GridSearchCV(
    estimator=svm,
    param_grid=param_grid,
    cv=5,
    scoring="accuracy"
)
```



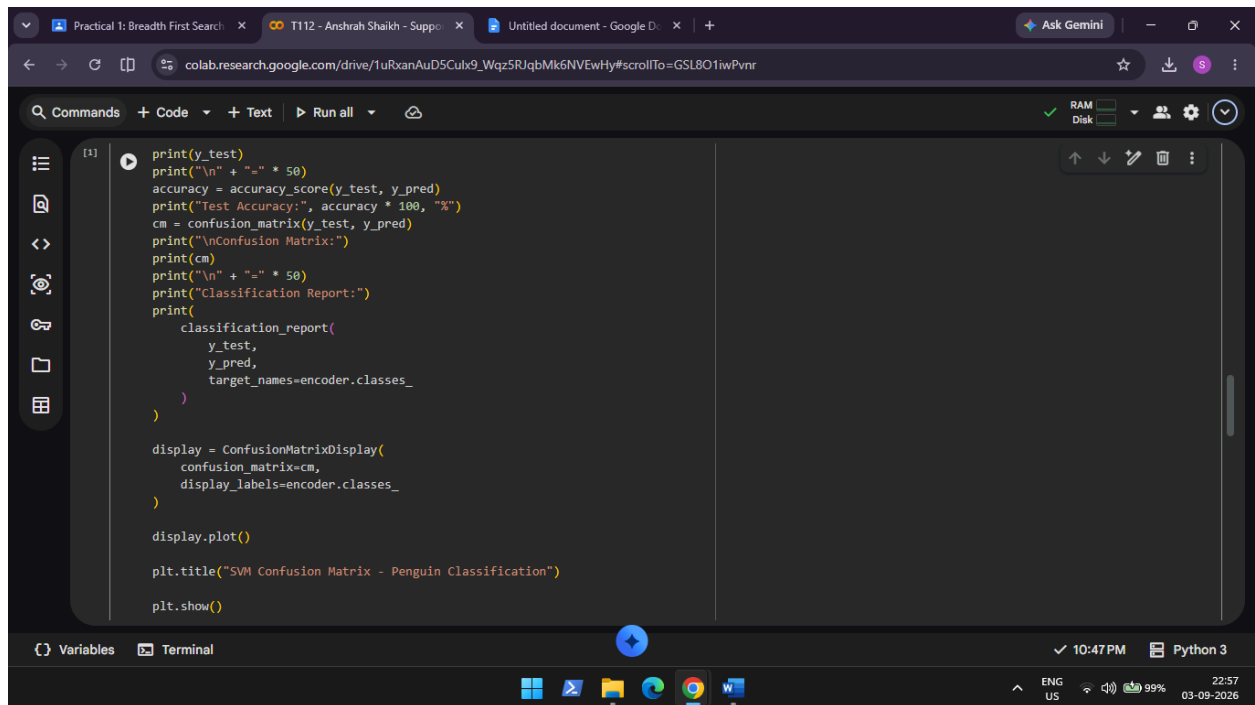
```
[1] grid_search.fit(X_train, y_train)

best_model = grid_search.best_estimator_

print("Best Parameters:")
print(grid_search.best_params_)
print("\nBest Cross-Validation Accuracy:")
print(grid_search.best_score_ * 100, "%")
print("\n" + "=" * 50)
y_pred = best_model.predict(X_test)
print("Predicted Results:")
print(y_pred)
print("\nActual Results:")
print(y_test)
print("\n" + "=" * 50)
accuracy = accuracy_score(y_test, y_pred)
print("Test Accuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
print("\n" + "=" * 50)
print("Classification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=encoder.classes_
    )
)
```

ANSHRAH SHAIKH  
T112  
TYCS  
AI PRACTICAL 5

# MVLU COLLEGE



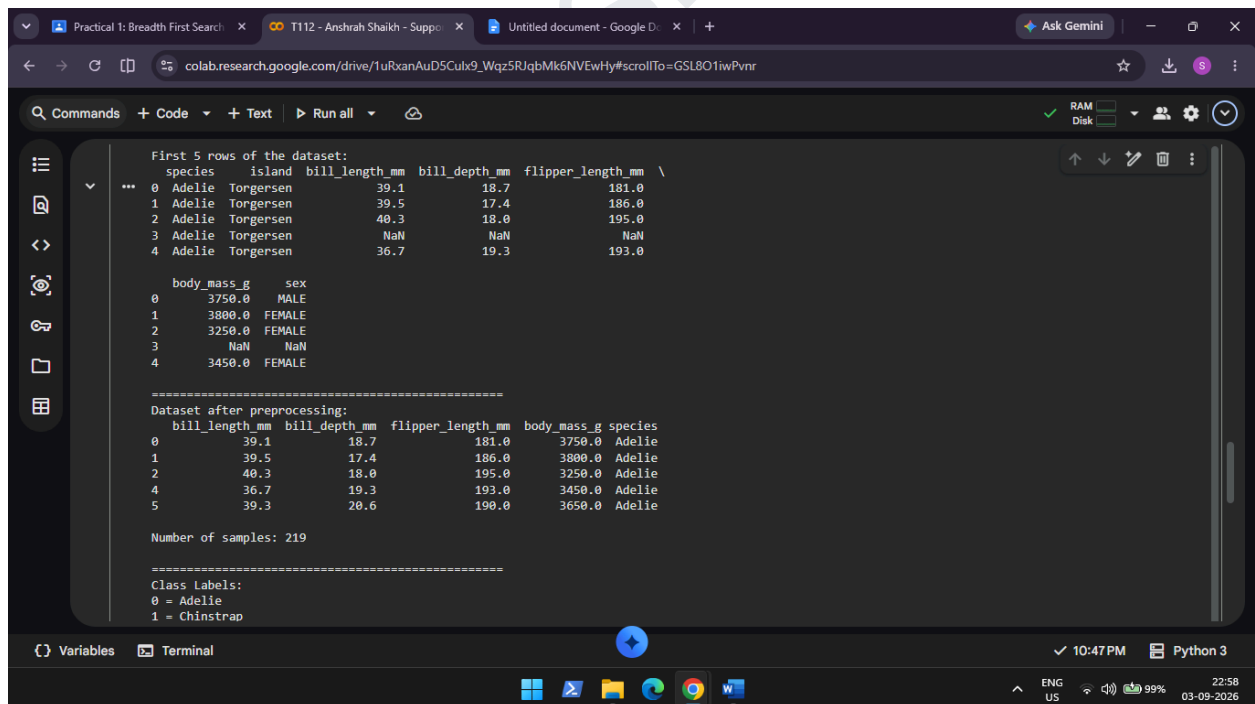
```
[1] print(y_test)
print("\n" + "=" * 50)
accuracy = accuracy_score(y_test, y_pred)
print("Test Accuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
print("\n" + "=" * 50)
print("Classification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=encoder.classes_
    )
)

display = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=encoder.classes_
)

display.plot()

plt.title("SVM Confusion Matrix - Penguin Classification")
plt.show()
```

OUTPUT:



```
First 5 rows of the dataset:
species  island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
0  Adelie  Tongersen      39.1          18.7          181.0
1  Adelie  Tongersen      39.5          17.4          186.0
2  Adelie  Tongersen      40.3          18.0          195.0
3  Adelie  Tongersen      NaN           NaN           NaN
4  Adelie  Tongersen      36.7          19.3          193.0

body_mass_g  sex
0      3750.0  MALE
1      3800.0  FEMALE
2      3250.0  FEMALE
3         NaN    NaN
4      3450.0  FEMALE

Dataset after preprocessing:
bill_length_mm  bill_depth_mm  flipper_length_mm  body_mass_g  species
0           39.1           18.7           181.0      3750.0  Adelie
1           39.5           17.4           186.0      3800.0  Adelie
2           40.3           18.0           195.0      3250.0  Adelie
4           36.7           19.3           193.0      3450.0  Adelie
5           39.3           20.6           190.0      3650.0  Adelie

Number of samples: 219

Class Labels:
0 = Adelie
1 = Chinstrap
```

ANSHRAH SHAIKH  
T112  
TYCS  
AI PRACTICAL 5

# MVLU COLLEGE

```
colab.research.google.com/drive/1uRxaAuD5Cux9_Wqz5RJqbMk6NVEwHy#scrollTo=GSL8O1wPvnr

Commands + Code + Text Run all

Best Parameters:
{'C': 10, 'gamma': 'scale', 'kernel': 'linear'}

Best Cross-Validation Accuracy:
97.71428571428572 %

Predicted Results:
[1 1 0 0 0 0 0 0 0 0 0 0 1 0 0 1 1 0 1 0 1 0 0 1 0 0 0 0 0 1 0 0 1 0 0 1 1
 0 0 0 1 0 1 0]

Actual Results:
[1 1 0 0 0 0 0 0 0 0 0 0 1 0 0 1 1 0 1 0 1 0 0 1 0 0 0 0 0 1 0 0 1 0 0 1 1
 0 0 0 1 0 1 0]

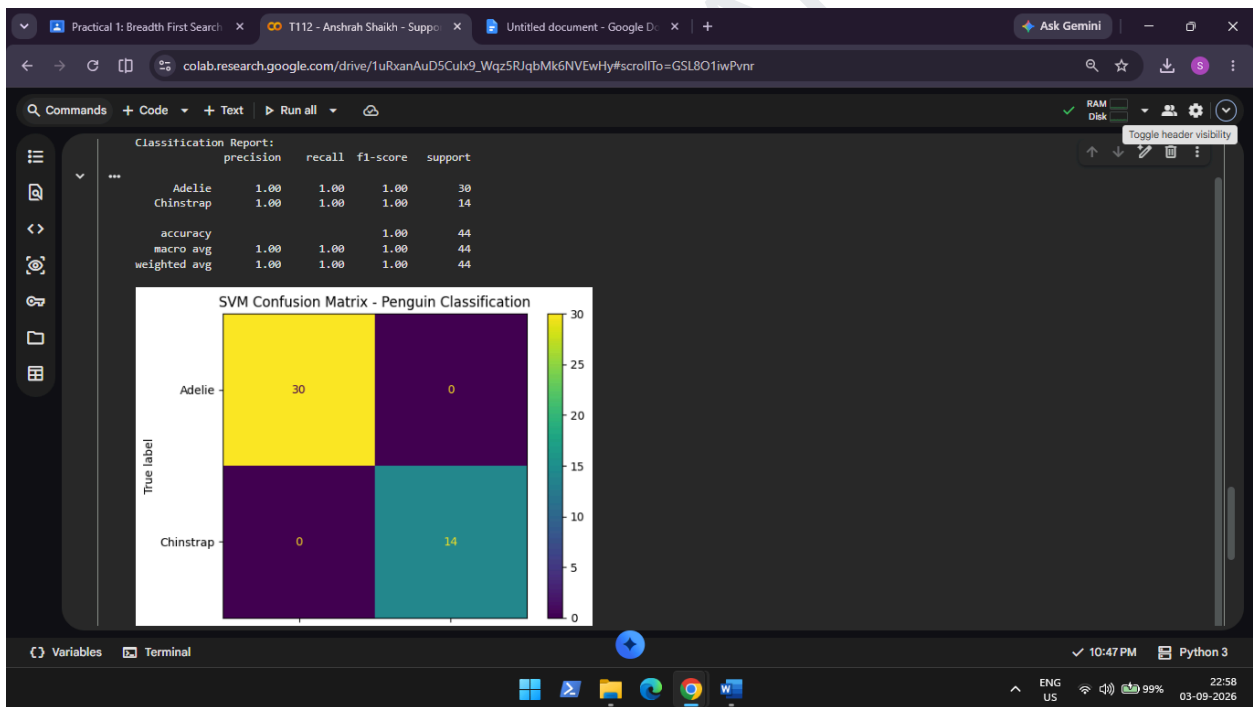
Test Accuracy: 100.0 %

Confusion Matrix:
[[30  0]
 [ 0 14]]

Classification Report:
              precision    recall  f1-score   support

   Adelie         1.00        1.00        1.00        30
  Chinstrap       1.00        1.00        1.00        14


```



ANSHRAH SHAIKH  
T112  
TYCS  
AI PRACTICAL 5